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10 May 1966

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MILITARY SPECIFICATION

ACTUATOR ASSEMBLY TEST SET

(For Torpedo Mk 46 Mod 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers requirements for the procurement of one type of Actuator Assembly Test Set (see 6.1.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

DRAWINGS

Bureau of Naval Weapons

DL 2540260	Actuator Assembly Test Set, and all documents listed thereon

FSC-1355

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PUBLICATIONS

Bureau of Naval Weapons

WD 28936	Special Manufacturing Procedure for Calibration of Actuator Assembly Test Set, DL 2540220

(Copies of drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Basic Requirements.

3.1.1 Contractor-furnished data. No data is required by this specification or by applicable documents referenced in Section 2, unless specified in contract or order (see 6.2).

3.1.2 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in manufacture of the Actuator Assembly Test Set shall be in accordance with requirements specified herein and with the applicable documents listed in BuWeps DL 2540260.

3.2 Performance and product characteristics.

3.2.1 Input power. The Actuator Assembly Test Set (hereafter referred to as test set) shall provide means for switching power on and off to the test set for the Actuator Assembly being tested. Indicator lamps shall be provided to ascertain that power is being applied. Four panel meters shall be provided on the test set front panel for measuring voltage and current levels.

Table I

AC Input Power

Voltage	Current
115 ±10 volts alternating current (v ac)	7.5 amperes maximum (amps max)

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3.2.2 Test set outputs. The test set shall include DC power supplies which shall provide -15 v dc at 10 ma and 42.5 v dc at 6.5 amps for the test set circuitry and external loads. The output load regulation shall be 10 millivolts maximum and ripple shall be 5 millivolts rms maximum. The voltages shall be accessible at specified terminals and shall be switched on and off using various combinations of the group switching.

3.2.3 Performance requirements. The test set shall provide the following test functions.

3.2.3.1 Resistance measurement. Switching functions shall place various inputs from the Actuator Assembly at test jacks (OHMS) on the test set panel. Resistance shall be measured with an external ohmmeter connected to the OHMS test jacks. The Actuator Assembly resistance to be measured shall be selected by test set switching as specified in table II.

Table II

Resistance Switching

Switch	Position	Function
FUNCTION	RESISTANCE	Switch Actuator Assembly inputs to MOTOR and INSTRUMENTATION Switches
RESISTANCE	INST, OFF, or MOTOR	Select output of INSTRUMENTATION switch or output of MOTOR switch for connection to OHMS test jacks.
INSTRUMENTATION	LOWER, EL, UPPER, or INST	Select actuator instrumentation circuit to be measured
MOTOR	LOWER, EL UPPER, or INST	Select actuator motor circuit to be measured
SHORT	UPPER, OFF, or LOWER	Short UPPER or LOWER positions of MOTOR switch

3.2.3.2 Isolation resistance. Placing the FUNCTION switch to ISOLATION and the ISOLATION switch to INT shall provide a 42 volt potential between the Actuator Assembly circuits and the Actuator Assembly frame. Current leakage from circuits to frame shall be indicated by a meter (ISOLATION CURRENT) on the test set panel. The meter shall indicate current in the

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range of zero to 20 microamperes. Placing the ISOLATION switch to EXT shall connect the actuator circuits and actuator frame across test jacks (MEGOHMS) on the test set panel, providing a capability for measuring isolation resistance with an external ohmmeter.

3.2.3.3 Motor drive control. The test set shall provide the following outputs to Actuator Assembly to control speed and direction of the actuator drive motors. Switches on the test set panel shall select the motor to be driven, speed, and the direction of drive.

NORMAL - CW: +36 ±0.5 volts at 0.7 ampere

NORMAL - CCW: -36 ±0.5 volts at 0.7 ampere

SLOW - CW: +10 ±1 volts at 0.1 ampere

SLOW - CCW: -10 ±1 volts at 0.1 ampere

3.2.3.4 Motor current measurement. A meter (MOTOR CURRENT) on the test set panel shall monitor the current of the motor drive outputs to the Actuator Assembly. The meter shall indicate current in the range of zero to 1000 milliamperes.

3.2.3.5 Motor time constant measurement. Test jacks (MOTOR OUTPUT) shall be provided on the test set panel for measurement of motor time constants. An external oscilloscope connected to the test jacks shall display time-versus-motor current of the motor driven by the test set.

3.2.3.6 Actuator position. A test fixture shall be provided which shall simulate the afterbody and fin shaft relationships to the Actuator Assembly. When mounted to the Actuator Assembly the test fixture shall set the position of the actuator and indicate actuator motor position and backlash. Backlash shall be indicated to nearest minute of arc.

3.2.3.7 Actuator rate measurement. Test jacks (RATE OUTPUT) shall be provided on the test set panel for measurement of actuator motor speed in degrees/second.

3.2.3.8 Continuity. Continuity from the test set through the Actuator Assembly connector shall be checked by the test set. A switch (CONTINUITY) shall be provided on the test set panel, which when on shall cause a continuity indicator to illuminate if the test set is connected to the Actuator Assembly, a condition which results in continuity between connector pins J1-M and J1-N.

3.2.3.9 Instrumentation voltage monitoring. A meter (INSTRUMENTATION VOLTAGE OUTPUT) on the test set panel shall monitor instrumentation voltage of the motor being driven by the test set. The INSTRUMENTATION VOLTAGE OUTPUT meter shall indicate voltage in the range -10 to zero to +10 volts.

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3.2.3.10 Test set -15 volt power supply monitor. A meter (15 V SOURCE) shall monitor the output of the -15 volt power supply. The 15 V SOURCE meter shall indicate in the range zero to 30 volts.

3.3 Workmanship. Workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with requirements of this specification. All parts used in the test set shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. Standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the test set shall be classified as follows:

4.2.1 Acceptance inspection.

(a) Acceptance Sampling (see 4.3.1)

(b) Acceptance Tests and Examinations (see 4.5)

4.3 Sampling.

4.3.1 Acceptance sampling. The tests and examinations specified in 4.5 shall be conducted on each test set to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the test set shall be subjected to tests under the following conditions:

(a)	Temperature	Room ambient (60 to 95 degrees F)
(b)	Humidity (Relative)	Room ambient, up to 95 percent
(c)	Altitude	Normal ground atmospheric pressure

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4.5 Acceptance tests and examinations. Unless otherwise specified each test set delivered under contract shall be subjected to the tests and examinations contained in 4.5.3.

4.5.1 Special notes.

4.5.1.1 Calibration. Calibration of the actuator assembly test set should be performed in accordance with WD 28936 and before conducting tests in 4.5.3.

4.5.1.2 Test sequence. Unless otherwise noted, sequence of testing procedures presented herein is mandatory and must be performed in the order shown.

4.5.1.3 Test results. Where space is provided in Appendix Report Form to indicate value or specific reading obtained, the specific reading shall be recorded if a limit or limits are given in Limits column for the test. If limits are not stated in Limits column, the individual performing the test or inspection shall initial in space provided in the Reading column indicating that requirements were met satisfactorily.

4.5.1.4 Acceptance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance, or when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units, which have not been corrected, will be considered for acceptance only upon formal application by the contractor to the Bureau of Naval Weapons via the cognizant Government representative.

4.5.2 Acceptance test equipment. Items of test equipment shown in table III are required to perform acceptance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals and procedures for and results of all calibrations shall be subject to approval of the cognizant Government Inspector. Items having equivalent characteristics may be substituted where necessary.

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Table III

Test Equipment

Item	Identification
Oscilloscope: (DC to 15 mc pass band; 23 nanosec rise time; 20 mv/cm deflection factor 6 cm by 10 cm display graticule)	Tektronix, Model 531 with 53/54 D plug-in preamplifier
Digital DC Voltmeter: (Range 0.001 to 999 v dc; accuracy: ±0.2% of reading, ±1 count)	Hewlett-Packard, Model 405 CR or 405 BR
Low Frequency Function Generator: (Frequency Range: 0.008 to 1200 cps in five decade ranges. Output Waveforms: Sinusoidal, square, and triangular; Maximum Output Voltage: At least 30 volts peak-to-peak across rated load (4000 ohms) for all three waveforms; Frequency Response: Constant within 0.2 db)	Hewlett-Packard, Model 202A
Resistor, 50 ohms, 50 watt, 1/2%. Resistor, 100 ohms, 50 watt, 1%. Resistor, 2.25 megohm, 1/2 watt, 1%	
Note: All test equipment used shall be isolated from a-c line ground.	

4.5.3 Test procedure.

4.5.3.1 Motor output test.

4.5.3.1.1 Verify that all power to the test set is off.

4.5.3.1.2 Connect a jumper between pin A and pin C of ACTUATOR INPUT connector. Connect a 50-ohm, 50-watt resistor across jumpered pins A, C, and pin B of ACTUATOR INPUT connector.

4.5.3.1.3 Position FUNCTION switch to ACTUATOR TEST and ACTUATOR switch to UPPER.

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4.5.3.1.4 Position 0-50 volts power supply line switch and OUTPUT switch to ON.

4.5.3.1.5 Position AC PWR switch to ON.

4.5.3.1.6 Position DC PWR switch to ON.

4.5.3.1.7 Connect digital DC voltmeter to -15 volt output. Digital DC voltmeter shall indicate -14.8 to -15.2 volts. Remove digital DC voltmeter from -15 volt output and connect it to 0 to +50 volt power supply. Set the power supply to indicate +42.4 to +42.6 volts on digital DC voltmeter.

4.5.3.1.8 Depress NORMAL switch and observe lamps within the NORMAL, (NORMAL) CW, and CCW switches. All three lamps shall light.

4.5.3.1.9 Connect digital DC voltmeter across 50-ohm, 50-watt resistor with negative lead going to pin B. Depress (NORMAL) CW switch. MOTOR CURRENT meter shall indicate 700 to 735 ma. Digital DC voltmeter shall indicate +35.5 to +36.5 volts.

4.5.3.1.10 Depress (NORMAL) CCW switch. MOTOR CURRENT meter shall indicate 700 to 735 ma. Digital DC voltmeter shall indicate -35.5 to -36.5 volts.

4.5.3.1.11 Depress the upper REVERSE switch. Observe lamps within the upper and lower switches. Both lamps shall light. Digital DC voltmeter shall indicate +35.5 to +36.5 volts.

4.5.3.1.12 Depress lower REVERSE switch. Digital DC voltmeter shall indicate -35.5 to -36.5 volts.

4.5.3.1.13 Position DC PWR switch to OFF.

4.5.3.1.14 Remove the 50-ohm, 50-watt resistor installed across jumpered pins A, C, and pin B of the ACTUATOR INPUT connector and replace with a 100-ohm, 50-watt resistor. Connect digital DC voltmeter across the 100-ohm, 50-watt resistor (negative lead goes to pin B).

4.5.3.1.15 Position DC PWR switch to ON.

4.5.3.1.16 Depress SLOW switch and observe lamps within the SLOW, (SLOW) CW, and CCW switches. All three lamps shall light.

4.5.3.1.17 Depress (SLOW) CW switch. Digital DC voltmeter shall indicate +9.0 to +11.0 volts.

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4.5.3.1.18 Depress (SLOW) CCW switch. Digital DC voltmeter shall indicate -9.0 to -11.0 volts.

4.5.3.1.19 Position DC PWR switch to OFF.

4.5.3.1.20 Remove jumper from pins A and C, and 100-ohm, 50-watt resistor from pin B on ACTUATOR INPUT connector.

4.5.3.1.21 Connect a 2.25-megohm, 1/2-watt resistor between pins C and V of J1.

4.5.3.1.22 Connect a jumper between pins N and M of ACTUATOR INPUT connector.

4.5.3.1.23 Position DC PWR switch to ON and position FUNCTION switch to ISOLATION. Hold ISOLATION switch on INT.

4.5.3.1.24 Observe ISOLATION CURRENT meter. Meter shall indicate 9 to 11 microamperes.

4.5.3.1.25 Position CONTINUITY switch to ON. Continuity indicator shall light.

4.5.3.1.26 Position DC PWR switch to OFF.

4.5.3.1.27 Remove 2.25-megohm, 1/2-watt resistor from pins C and V and remove jumper between pins N and M, all of ACTUATOR INPUT connector.

4.5.3.2 Rate output test.

4.5.3.2.1 Position oscilloscope controls as follows:

Control	Position
TRIGGER MODE	AC SLOW
TRIGGER SLOPE	EXT (-)
TRIGGER STABILITY	FULLY CW
TRIGGER LEVEL	FULLY CW
TIME/CM	100 MILLISEC
5X MAGNIFIER	OFF
MULTIPLIER	1
MILLIVOLTS/CM	100

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Control	Position
MV/CM MULTIPLIER	2
VARIABLE	CALIBRATED (FULLY CW)
INPUT MODE	A, DC

4.5.3.2.2 Connect low frequency function generator (signal generator) between pins C and D of TEST FIXTURE INPUT connector located on the test set panel. Pin C connects to upper output terminal, pin D connects to lower output terminal. Do not connect ground strap.

4.5.3.2.3 Connect input A of oscilloscope to upper output of signal generator. Connnect oscilloscope ground to lower output terminal. Connect the SYNC OUT from signal generator to TRIGGER input of oscilloscope. Connect SYNC output ground to lower output terminal of signal generator.

4.5.3.2.4 Set signal generator waveform to triangular and frequency to 0.5 cps. Set amplitude so that peak-to-peak value of waveform is approximately 5 cm as indicated on oscilloscope.

4.5.3.2.5 Position DC PWR switch to ON.

4.5.3.2.6 Turn TRIGGER STABILITY control CCW until oscilloscope trace no longer triggers. Turn TRIGGER LEVEL control until a trigger is obtained. Trigger should occur as triangular wave begins positive-going half cycle. If this does not occur reverse all leads connected to signal generator output.

4.5.3.2.7 Adjust signal generator amplitude and frequency controls and position oscilloscope vertical and horizontal controls to obtain a trace slope on the graticule at 1 volt/second. Trace shall slope from left at 0 cm to right at 5 cm.

4.5.3.2.8 Connect input A of oscilloscope to a ground terminal on oscilloscope. Position MV/CM MULTIPLIER to 1. Position trace on top line of graticule.

4.5.3.2.9 Connect input A of oscilloscope to RATE OUTPUT terminals on test panel.

4.5.3.2.10 A negative-going trace shall appear on oscilloscope. (See figure 1.) Final value shall be at 6 cm below top line.

4.5.3.2.11 Position AC PWR and DC PWR switches to OFF.

4.5.3.2.12 Disconnect oscilloscope and signal generator from test set.

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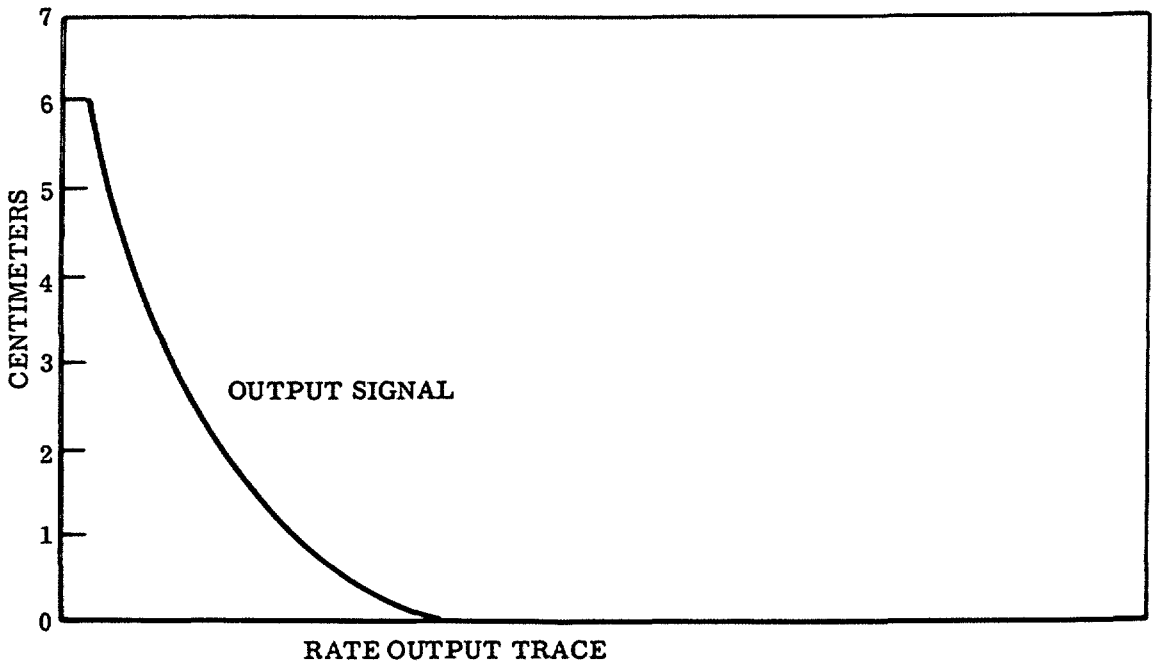


Figure 1. Rate Output

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4.5.4 Visual inspections.

4.5.4.1 Identification marking (major).

4.5.4.1.1 The nomenclature, control callout and identification data shall be correct, clear and legible. (Reference: BuWeps DL 2540260.)

4.5.4.2 Interchangeability.

4.5.4.2.1 The connector receptacles shall be correct and capable of accepting the mating units without resorting to selection, modification or fitting. (Reference: BuWeps DL 2540260.)

4.5.5 Process verification.

4.5.5.1 Criteria stipulated in this paragraph are applicable to the production process instead of specific units of product. Inspections stipulated are general in scope and technique and frequency of verification will be contingent upon specific process conditions. Positive action by the contractor is required when process criteria findings indicate poor inspection by the contractor.

4.5.5.2 Lacing (minor).

4.5.5.2.1 Lacing and cable configuration shall be in accordance with procedures approved by the procuring activity. Specific method of verifying the adequacy of cable lacing shall include visual and dimensional verification.

4.5.5.3 Soldering (minor).

4.5.5.3.1 Soldered joints shall be in accordance with good commercial practices. Unsoldered connections or those of a "cold" solder or "rosin" joint shall be considered defective.

4.5.5.4 Cleanliness (major).

4.5.5.4.1 Presence of foreign material in assembly shall classify unit as defective. Foreign material includes anything not intended to be present.

4.5.6 Packaging and packing inspections.

4.5.6.1 Packaging.

4.5.6.1.1 Level C packaging (see 5.1.3) shall comply with commercial practice.

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4.5.6.2 Packing.

4.5.6.2.1 Level C packing shall be used (see 5.2.3). The test set shall be protected against damage during direct shipment from supplier to the first receiving activity for immediate use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level C as specified (see 6.2b and 6.4).

5.1.1 Level A. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation and packaging shall be in accordance with the supplier's commercial practices when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level C as specified (see 6.2c and 6.4).

5.2.1 Level A. Not applicable.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The test set, packaged as specified in 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

6. NOTES

6.1 Intended use and description.

6.1.1 Intended use. The test set is intended for use in a manufacturing facility for acceptance testing of Actuator Assembly DL 2540014.

6.1.1.1 The test set is expected to operate in the following environments:

(a) Temperature. Room ambient temperature between the limits of 65° to 95°F (18.33° to 35°C).

(b) Humidity (relative). Room ambient relative humidity not to exceed 95 percent.

(c) Atmospheric pressure. Prevailing room ambient.

(d) Vibration. None.

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(e) Shock. None.

(f) Acceleration. None.

6.1.2 Description.

6.1.2.1 Mechanical. The test set is a rectangular cabinet containing a rack mounted test panel and a rack mounted power supply.

6.1.2.2 Electrical. The test set simulates the electrical inputs and outputs of the Actuator Assembly.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title number and date of this specification.

(b) Required level of preservation and packaging.

(c) Required level of packing.

(d) Required data.

(e) Any exceptions to this specification.

(f) All additional samples required shall be provided and inspected at the expense of the supplier when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.3 Definitions. For the purpose of this specification the following have been established.

6.3.1 Operating. Operating is the condition of the test set performing its required function.

6.3.2 Temperature. The temperature of the environment to which the test set may be exposed, expressed in degrees Fahrenheit (°F).

6.3.3 Pressure. The pressure of the environment to which the test set may be exposed, expressed in pounds per square inch absolute (psia).

6.3.4 Humidity. The relative humidity of the environment to which the test set is exposed, expressed as a percent.

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6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A	Not applicable.
Level B	Not applicable.
Level C	This level shall be used when the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

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REPORT FORM

QUALITY ASSURANCE PROVISION

ACCEPTANCE TESTS AND EXAMINATIONS

FOR

ACTUATOR ASSEMBLY TEST SET

DL 2540260

(For Torpedo Mk46 Mod 1)

EQUIPMENT STATUS	
CONTRACTOR CERTIFICATION	GOVERNMENT ACCEPTANCE
Contractor	Date of Acceptance
Contract	Accepted by
Actuator Assembly Test Set Serial No.	Title
Insp. Req. Satisfied	Agency
Test Fixture Serial No.	

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APPENDIX

REPORT FORM

QUALITY ASSURANCE PROVISION

ACCEPTANCE TEST AND EXAMINATION

ACTUATOR ASSEMBLY TEST SET

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the acceptance tests and examinations of 4.5 of this specification. When completed with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the test set bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. The acceptance tests and examinations of 4.5 shall be completed and the results recorded or verified in this report form prior to acceptance of the test set. Unless otherwise noted the sequence of testing, recording and verification, as performed in 4.5 is mandatory and must be performed in the order shown.

20.2 Recording of results. Where space is provided to indicate the value of specific reading obtained, the specific reading will be recorded if a limit or limits is given in the limit column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limit column, the individual performing the test or inspection shall initial in the space provided in the Reading column indicating that the requirement has been met satisfactorily.

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30. ACCEPTANCE TEST AND EXAMINATION

	Reference	Limits	Reading
30.1 Test equipment.	4.5.2	In Calibration	_____
30.2 Test procedure.			
30.2.1 Motor output test.	4.5.3.1		
-15 volt source	4.5.3.1.7	-14.8 to -15.2 volts	_____
+42.5 volt source		+42.4 to +42.6 volts	_____
NORMAL switch lamp	4.5.3.1.8	Shall light	_____
(NORMAL) CW switch lamp	4.5.3.1.8	Shall light	_____
(NORMAL) CCW switch lamp	4.5.3.1.8	Shall light	_____
MOTOR CURRENT meter	4.5.3.1.9	700 to 735 ma	_____
Digital DC Voltmeter	4.5.3.1.9	+35.5 to +36.5 volts	_____
MOTOR CURRENT meter	4.5.3.1.10	700 to 735 ma	_____
Digital DC Voltmeter	4.5.3.1.10	-35.5 to -36.5 volts	_____
Upper REVERSE switch	4.5.3.1.11	Shall light	_____
Lower REVERSE switch	4.5.3.1.11	Shall light	_____
Digital DC Voltmeter	4.5.3.1.11	+35.5 to +36.5 volts	_____
Digital DC Voltmeter	4.5.3.1.12	-35.5 to -36.5 volts	_____
SLOW switch lamp	4.5.3.1.16	Shall light	_____
(SLOW) CW switch lamp	4.5.3.1.16	Shall light	_____

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30.2.1 (Continued)	Reference	Limits	Reading
(SLOW) CCW switch lamp	4.5.3.1.16	Shall light	_____
Digital DC Voltmeter	4.5.3.1.17	+9.00 to +11.0 volts	_____
Digital DC Voltmeter	4.5.3.1.18	-9.00 to -11.0 volts	_____
ISOLATION CURRENT meter	4.5.3.1.24	9 to 11 micro-amperes	_____
Continuity indicator	4.5.3.1.25	Shall light	_____
30.2.2 Rate output test.	4.5.3.2		
Rate Output	4.5.3.2.10	6 cm amplitude	_____
30.3 Visual inspections.	4.5.4		
30.3.1 Identification marking (major)	4.5.4.1		
Nomenclature, Control and identification callouts	4.5.4.1.1		_____
30.3.2 Interchangeability.	4.5.4.2		
Connector receptacles	4.5.4.2.1		_____
30.4 Process verification.	4.5.5		
30.4.1 Lacing (minor).	4.5.5.2		
Lacing and cable configuration	4.5.5.2.1		_____
30.4.2 Soldering (minor).	4.5.5.3		
Soldered joints	4.5.5.3.1		_____

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	Reference	Limits	Reading
30.4.3 Cleanliness (major).	4.5.5.4		
Foreign material	4.5.5.4.1		_____
30.5 Packaging and packing instructions.	4.5.6		
30.5.1 Packaging.	4.5.6.1		
Level C packaging	4.5.6.1.1		_____
30.5.2 Packing.	4.5.6.2		
Level C packing	4.5.6.2.1		_____

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-A-82451(OS) ACTUATOR ASSEMBLY TEST SET (FOR TORPEDO MK 46 MOD 1)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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C-8378

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DEPARTMENT OF THE NAVY
Naval Ordnance Systems Command
Washington, D.C. 20360

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